Huawei Access Node ELEC-ACC-57 QoS Congestion Event Resolution SOP

Node: ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9797

Description

The failure domain 'QoS / Congestion Event' on the Huawei Access Node ELEC-ACC-57 at Electronic City Hub, is primarily attributed to the occurrence of microbursts and interface tail drops, which are indicative of QoS queue starvation. This phenomenon arises due to the insufficiency of buffering capacity on the egress interface, causing packets to be dropped and leading to a congestion event. Additionally, the Huawei Access Node's QoS policy configuration may be inadequately set up, resulting in incorrect QoS queue configurations, and therefore, insufficient buffer allocation. Furthermore, the possibility of misconfigured queue scheduling algorithms or incorrect queue weighting factors cannot be ruled out. The likelihood of these events occurring is heightened in scenarios where the access node is experiencing high traffic load or is being subjected to DoS/DDoS attacks.

Pre-requisite Checks

- Physical presence at the Electronic City Hub site to access the Huawei Access Node ELEC-ACC-57.

- Verify the redundant link status to ensure it is operational and can take over in case of failure.

- Ensure all personnel involved are familiar with the Huawei CLI dialect and the access node's configuration.

Potential Root Causes

Cause	Probability
Inadequate buffering capacity on egress interface	High
Misconfigured QoS policy	Medium
Incorrect queue scheduling algorithm or weighting factor	Low

Diagnostic Commands & Expected Outputs

```
Command: display interface e1/0/1, Expected Output: Interface e1/0/1 is up and operational., Description: This command displays the interface status, helping to identify whether the egress interface is up and operational.

Command: display qos queue e1/0/1, Expected Output: QoS queue configurations for egress interface e1/0/1 are correctly set., Description: This command displays the QoS queue configurations for the egress interface, helping to identify whether the queue configurations are correctly set or not.

Command: display buffer e1/0/1, Expected Output: Buffering capacity on egress interface e1/0/1 is sufficient., Description: This command
```

```
displays the buffering capacity on the egress interface, helping to
identify whether the buffering capacity is sufficient or not.
```

Resolution Steps

Step 1: Step: Increase buffering capacity on egress interface, Command: buffer e1/0/1 size 32768, Description: Increase the buffering capacity on the egress interface to prevent packet drops and congestion events.

Step 2: Step: Reconfigure QoS policy, Command: qos queue e1/0/1 priority 0 weight 10, Description: Reconfigure the QoS policy to allocate sufficient buffer allocation to the egress interface and prevent QoS queue starvation.

Step 3: Step: Verify queue scheduling algorithm and weighting factor, Command: display qos queue e1/0/1, Description: Verify the queue scheduling algorithm and weighting factor to ensure they are correctly set and not contributing to the congestion event.

Post-Check Verification

```
Command: display interface e1/0/1, Expected Output: Interface e1/0/1
is up and operational., Description: Verify the interface status to
ensure it is up and operational after the resolution steps.

Command: display qos queue e1/0/1, Expected Output: QoS queue
configurations for egress interface e1/0/1 are correctly set.,
Description: Verify the QoS queue configurations to ensure they are
correctly set after the resolution steps.

Command: display buffer e1/0/1, Expected Output: Buffering capacity on
egress interface e1/0/1 is sufficient., Description: Verify the
buffering capacity on the egress interface to ensure it is sufficient
and not contributing to the congestion event.
```

Roll-back Procedures

- Step: Reset buffering capacity on egress interface, Command: buffer e1/0/1 reset, Description: Reset the buffering capacity on the egress interface to its original value if the fix fails.

- Step: Restore original QoS policy, Command: qos queue e1/0/1 restore, Description: Restore the original QoS policy if the fix fails.